FIITJEE

ALL INDIA TEST SERIES

FULL TEST – VIII

JEE (Main)-2023

TEST DATE: 25-03-2023

ANSWERS, HINTS & SOLUTIONS

Physics

PART – A

SECTION – A

1. C

Sol. $r_C = \frac{v^2}{a_\perp} = \frac{v^2}{\left(\frac{Bqv\sin\theta}{m}\right)} = \frac{mv}{Bq\sin\theta}$

2. D

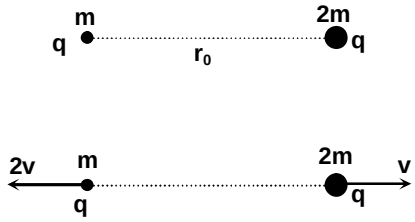
Sol. Applying conservation of energy

$$\frac{kq(q)}{r_0} = \frac{1}{2}m(2v)^2 + \frac{1}{2}(2m)v^2$$

$$\frac{kq^2}{r_0} = 3mv^2$$

$$v = \sqrt{\frac{kq^2}{3mr_0}}$$

$v_P = 2v$



3. A

Sol. $L + e = \frac{7\lambda}{4}$

$$k = \frac{2\pi}{\lambda} = \frac{2\pi}{\frac{4}{7}(L+e)} = \frac{7\pi}{2(L+e)}$$

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4. A

Sol. $C_{eq_{AB}} = C_1 = 3C_0 + \frac{2C_0}{3} = \frac{11C_0}{3}$

$C_2 = C_0 + \frac{6C_0}{5} = \frac{11C_0}{5}$

$C_3 = 2C_0 + \frac{3C_0}{4} = \frac{11C_0}{4}$

$C_1 : C_2 : C_3 = \frac{1}{3} : \frac{1}{5} : \frac{1}{4} = 20 : 12 : 15$

5. C

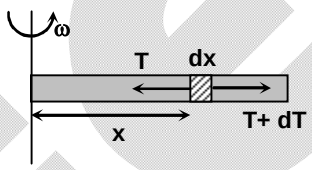
Sol. $-dT = dm\omega^2 x$

$-\int_0^T dT = \int_L^{L/2} \lambda_0 x\omega^2 x dx$

$T = \frac{\lambda_0\omega^2}{3}\left[x^3\right]_{L/2}^{L} = \frac{7\lambda_0\omega^2 L^3}{24}$

Also, $M = \int_0^L \lambda_0 x dx$

$M = \frac{\lambda_0 L^2}{2}$



6. B

Sol. $\left|V_P - V_Q\right| = \frac{B\omega(r+d)^2}{2} - \frac{B\omega r^2}{2} = \frac{B\omega}{2}\left[(r+d)^2 - r^2\right] = B\omega rd$

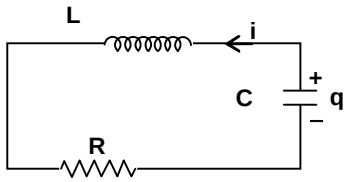
7. D

Sol. Apply KVL

$iR - L\frac{di}{dt} - \frac{q}{C} = 0$

$L\frac{d^2q}{dt^2} + \frac{Rdq}{dt} + \frac{q}{C} = 0$

$\therefore\ Q_0 e^{-\frac{Rt}{2L}} \cos\left(\sqrt{\frac{1}{LC} - \frac{R^2}{4L^2}}\, t\right)$



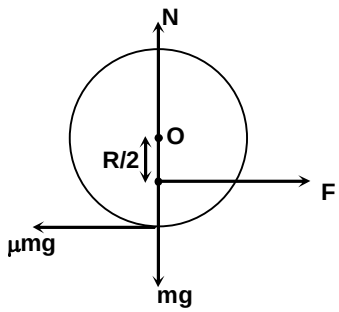
8. A

Sol. $\alpha = 0 \Rightarrow \tau_0 = 0$

$\frac{fR}{2} - \mu mgR = 0$

$F = 2\mu mg$

$a = \frac{F - \mu mg}{m} = \mu g$



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9. C

Sol. volume remains conserved

$\therefore\ 4\pi R^2 d = \frac{4}{3}\pi r^3 \Rightarrow r = \left(3R^2 d\right)^{1/3}$

Ratio = $\left(\frac{4T}{R}\right)\frac{r}{2T} = \frac{2}{R}(3R^2 d)^{1/3} = 2\left(\frac{3d}{R}\right)^{1/3}$

10. A

Sol. As block slides upward,

$\therefore f = \mu N$

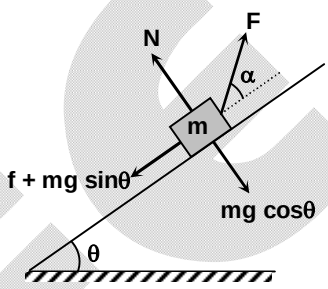
$F\cos\alpha = mg\sin\theta + f$

$N = mg\cos\theta - F\sin\alpha$

$\therefore F\cos\alpha = mg\sin\theta + \mu(mg\cos\theta - F\sin\alpha)$

$F = \frac{mg(\sin\theta + \mu\cos\theta)}{(\cos\alpha + \mu\sin\alpha)}$

$\Rightarrow F_{min} = mg\frac{\left[\sin\theta + \mu\cos\theta\right]}{\sqrt{1+\mu^2}}$



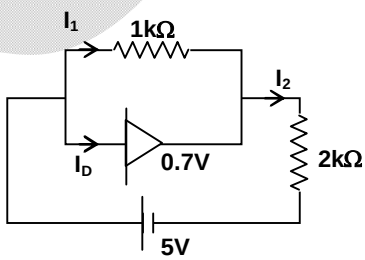
11. B

Sol. As the diode is not ideal $\Rightarrow I_1 \neq 0$

$I_1 + I_D = I_2$

$\frac{0.7V}{1\,k\Omega} + I_D = \frac{4.3V}{2k\Omega}$

$\therefore I_D = 2.15\ mA - 0.7\ mA = 1.45\ mA$



12. A

Sol. $\Delta Q = mS\Delta T$

and $\ell = \ell_0\alpha\Delta T$

Also, $\ell_0 = \frac{V}{A} = \frac{m}{dA}$

$\Delta Q = \frac{mS\ell Ad}{\alpha m} = \frac{dAs\ell}{\alpha}$

13. C

Sol. $Q = W + \Delta U$

$= \frac{1}{2}\left[P_0 + 2P_0\right]2V_0 + (1)\left(\frac{3R}{2}\right)\left[\frac{6P_0V_0}{R} - \frac{P_0V_0}{R}\right] = 1C_{avg}\left(\frac{5P_0V_0}{R}\right)$

14. D

Sol. $\left(\frac{\varepsilon}{r+1}\right)^2(1) = \left(\frac{\varepsilon}{r+9}\right)^2 9$

$\Rightarrow r = 3\Omega$

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15. D

Sol. $\delta_{min} = A = 2i - A \Rightarrow i = A$

$\therefore \sin i = n \sin r$

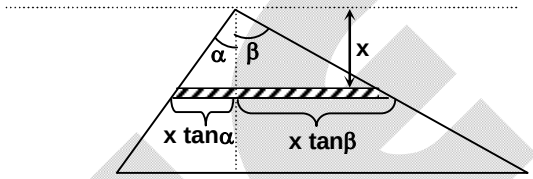
$\text{Sin } A = n\sin\left(\frac{A}{2}\right)$

$\Rightarrow A = 2\cos^{-1}\left(\frac{n}{2}\right)$

16. B

Sol. $I = \int dmx^2 = \int_0^h \sigma x(\tan\alpha + \tan\beta)x^2 dx$

$= \frac{m(\tan\alpha + \tan\beta)}{\frac{1}{2}h(\tan\alpha + \tan\beta)h}\frac{h^4}{4} = \frac{mh^2}{2}$



17. D

Sol. $f = \frac{1}{2\pi\sqrt{L_{eq}C_{eq}}}$

$L_{eq} = L + \frac{3L(6L)}{9L} = 3L$ and $C_{eq} = C$

$\therefore f = \frac{1}{2\pi\sqrt{3LC}}$

18. D

Sol. 3rd excitation potential = $\frac{E_4 - E_1}{e}$

$= \frac{1}{e}(2)^2(13.6eV)\left[1 - \frac{1}{16}\right] = 51V$

19. C

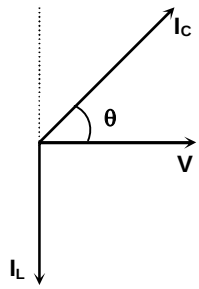
Sol. The Phasor diagram of the circuit is

If I and V are in same phase

$\therefore I_L = I_C \sin\theta$

$\frac{V}{x_L} = \frac{VR}{\left(R^2 + x_C^2\right)}$

$x_L = \frac{R^2 + x_C^2}{R} = \frac{4^2 + 3^2}{4} = \frac{25}{4}\Omega$



20. A

Sol. Mixing of two gases is a spontaneous process

SECTION – B

21. 1

Sol. $F = v_r\frac{\Delta m}{\Delta t} = (100)\left(\frac{10}{1000}\right)\frac{60}{60}N = 1N$

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22. 3

Sol. $\frac{1}{2}mv_e^2 - \frac{3}{2}\frac{GMm}{R} = 0$

$v_e = \sqrt{\frac{3GM}{R}}$

23. 81

Sol. $W = \int_{(0,0)}^{(3,9)} (x^2 y dx + \frac{x^3}{3} dy) = \left[\frac{x^3 y}{3}\right]_{(0,0)}^{(3,9)} = 81 \text{ J}$

24. 100

Sol. $K_\alpha = \left(\frac{A-4}{A}\right)Q \Rightarrow 48 = \frac{A-4}{A}(50)$

$\Rightarrow A = 50$

25. 4

Sol. $\theta = \cos^{-1}\left(\frac{A}{2A}\right) + \cos^{-1}\left(\frac{A\sqrt{3}}{2A}\right) = \frac{\pi}{2}$

$t = T/4$

26. 6

Sol. $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$, also $m = \frac{v}{u}$

$\therefore\ m = \frac{f}{f+u}$

$\therefore\ \frac{m_1}{m_2} = \left(\frac{20}{20-25}\right)\left(\frac{20-50}{20}\right) = 6$

27. 48

Sol. $H = 5(t+0.2)^2$

$H - 6 = 5t^2$

$H \approx 48$ m

28. 78

Sol. $\frac{D}{d}t(\mu - 1) = \frac{13}{2}\frac{\lambda D}{d}$

$t\left(\frac{3}{2} - 1\right) = \frac{13}{2} \times 6000\ \text{Å}$

$\therefore\ t = 78 \times 10^{-7}$ m

29. 9

Sol. $\%\text{ error in } \left(\frac{(a+b)c}{d}\right) = \left[\frac{\Delta(a+b)}{(a+b)} + \frac{\Delta c}{c} + \frac{\Delta d}{d}\right] \times 100$

$= \left(\frac{0.6}{30} + \frac{0.1}{10} + \frac{0.2}{10}\right) \times 100 = 5\%$

30. 8

Sol. $s = \int_0^2 (2t - t^2)dt + \int_2^4 (t^2 - 2t)dt = 8m$

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Chemistry

PART – B

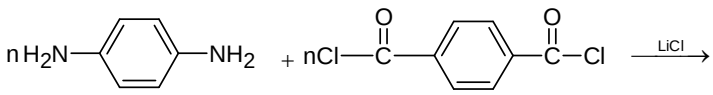
SECTION – A

31. A

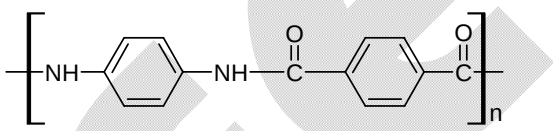
Sol. Ketose sugar like fructose gives deep red colour on reaction with conc. HCl and resorcinol (Seliwanoff's reagent)

32. D

Sol.



1,4-phenylene diamine terephthaloyl chloride



33. D

Sol. $ZnO(s) \rightleftharpoons Zn^{2+} + \frac{1}{2}O_2(g) + 2e$

Due to anion deficiency, F-centre is formed.

34. D

Sol. XeF_2, Xe is sp^3d hybridised

XeO_3, Xe is sp^3 hybridised

$XeOF_4$, Xe is sp^3d^2 hybridised

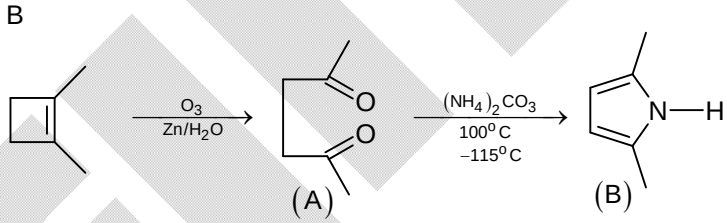
$\overset{\oplus}{N}O_2$, N is sp hybridized

35. D

Sol. In solid state $BeCl_2$ is polymeric but $AlCl_3$ is dimeric.

36. B

Sol.



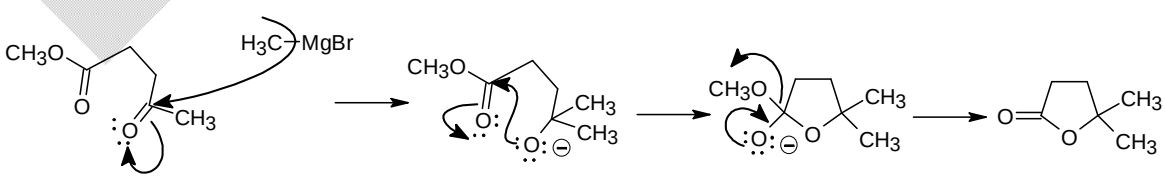
B is heterocyclic aromatic compound containing five membered ring.

37. D

Sol. Basic strength of II and III decreases due to ortho effect.

38. D

Sol.



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39. A

Sol. Gabriel's phthalimide synthesis involves the S_N2 reaction on an alkyl halide. S_N2 on aromatic halide is not possible.

40. B

Sol. 25 ml 0.04 M $Na_2CO_3 = 25 \times 0.04 = 1$ milimole Na_2CO_3

20 ml 0.05 M $HCl = 20 \times 0.05 = 1$ milimole HCl

$$Na_2CO_3 + HCl \longrightarrow NaHCO_3 + NaCl$$
$$1 \qquad 1 \qquad 1$$

Resulting solution contains 1 milimole of $NaHCO_3$

$$pH = \frac{pK_{a_1} + pK_{a_2}}{2} = \frac{6.4 + 10.3}{2} = 8.35$$

41. D

Sol. Lesser be the standard reduction potential greater be tendency to act as a reducing agent.

42. B

Sol. $e_g --$

$CFSE = -2.4\Delta_O + 3P$

t_{2g} 

43. A

Sol. $$K_2Cr_2O_7 + 4KCl + 3H_2SO_4 \longrightarrow \underset{(A)}{2CrO_2Cl_2} + 3K_2SO_4 + 3H_2O$$

$$CrO_2Cl_2 + 4NaOH \longrightarrow Na_2CrO_4 + 2NaCl + 2H_2O$$

Number of atoms in $CrO_2Cl_2 = 2$.

44. D

45. C

Sol. For an adiabatic process,

$$\frac{P}{P'}\left(\frac{V}{V'}\right)^{\gamma} = 1$$

or $$32\left(\frac{d'}{d}\right)^{\gamma} = 1$$

so, $\frac{d}{d'} = 128 \qquad (\gamma = 7/5)$

46. D

Sol. $$PCl_3 + 3H_2O \longrightarrow 3HCl + H_3PO_3$$

$$H_3PO_3 + 2AgNO_3 + H_2O \longrightarrow H_3PO_4 + 2Ag + 2HNO_3$$

47. A

Sol. Factual

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48. B

Sol. 3S orbital has two radial nodes.

49. C

Sol. $\frac{d\ell nk}{dT} = \frac{E_a}{RT^2}$

Greater be activation energy, greater be the temperature dependency of reaction rate.

50. C

Sol. Volume of one drop $= \frac{2}{35}$ ml.

Number of moles in one drop $= \frac{2 \times 1.2}{35 \times 70} = \frac{12}{(35)^2}$

Number of molecules in one drop $= \frac{1.2}{(35)^2} N_A$

SECTION – B

51. 2

Sol. Mean free path

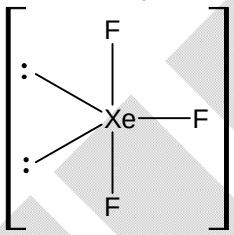
$\lambda = \frac{1}{\sqrt{2}\pi\sigma^2 \bar{N}}$ $\quad \bar{N} \propto \left(\bar{N} = \text{number of molecules per unit volume}\right)$

i.e. $\lambda \propto \frac{T}{P}$

If temperature is increased by 4 times and pressure by 2 times, λ will be increased by 2 times.

52. 2

Sol. $XeF_4 + 36F_5 \longrightarrow XeF_3^+ + 36F_6^-$



2 lone pairs

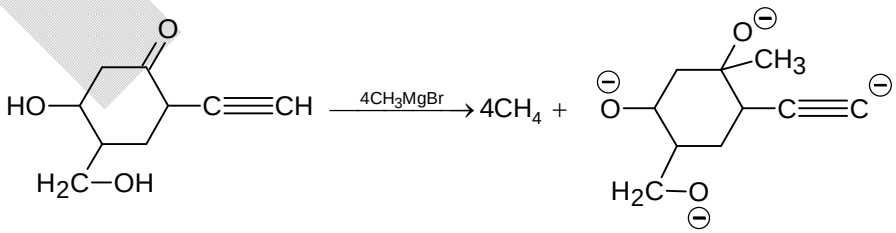
53. 3

Sol. In NaCl, Na^+ ions are present in the octahedral voids. The total number of Na^+ = 4. If the Na^+ ions along one axis joining the faces are removed, the Na^+ at the body centred position is removed.

i.e. Number of Na^+ = 4 – 1 = 3.

54. 4

Sol.



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55. 8

Sol. Total number of double bonds and chiral centres = 3.

i.e. total number of stereoisomers $= 2^3 = 8$.

56. 93

Sol. $\Delta T_f = K_f \dfrac{w_B \times 1000}{m_B \times w_A}$

$2.8 = 1.86 \times \dfrac{w_B \times 1000}{62 \times 1000}$

$w_B = 93$

57. 8

Sol. $\overset{+1}{Cu}_2\overset{-2}{S} + 2O_2 \longrightarrow 2\overset{+2}{Cu}O + \overset{4}{S}O_2$

n-factor $= |2-1| \times 2 + |4-(-2)|$

$= 2 + 6 = 8$

58. 1

Sol. $O_3(g) \overset{K_{eq}}{\rightleftharpoons} O_2(g) + O(g)$

$\dfrac{[O_2][O]}{[O_3]} = K_{eq} \Rightarrow [O] = K_{eq}\dfrac{[O_3]}{[O_2]}$

Rate determining step

$NO_2(g) + O(g) \xrightarrow{k} NO_3(g)$

rate $= k[NO_2][O] = k[NO_2]K_{eq}\dfrac{[O_3]}{[O_2]}$

$= k\ K_{eq}[NO_2][O_3][O_2]^{-1}$

$= k'[NO_2][O_3][O_2]^{-1}$

$(k' = K_{eq}.k)$

Overall order = 1 + 1 – 1 = 1.

59. 6

Sol. $Ca_3P_2 + 6H_2O \longrightarrow 3Ca(OH)_2 + 2PH_3$

1 mole of Ca_3P_2 gives 2 moles of PH_3

$\therefore$ 3 moles of Ca_3P_2 gives 6 moles of PH_3

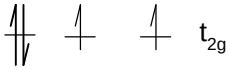
60. 2

Sol. $[CoF_6]^{3-}$ oxidation state of Co is +3.

Co has d^6 electronic configuration



e_g



t_{2g}

F^- is weak field ligand. Number of unpaired electrons in $t_{2g} = 2$.

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Mathematics

PART – C

SECTION – A

61. A

Sol. $D \geq 0$

$4a^2 - 36 \geq 0$

$a \in (-\infty, -3] \cup [3, \infty)$ …(i)

$-6 < -\frac{2a}{2} < 1$

$-1 < a < 6$ …(ii)

$f(-6) > 0$ $f(1) > 0$

$36 - 12a + 9 > 0$ $1 + 2a + 9 > 0$

$a < \frac{45}{12} = \frac{15}{4}$ …(iii) $a > -5$

Eq. (i), (ii) and (iii)

$a \in \left[3, \frac{15}{4}\right)$

62. B

Sol. $\frac{2n+1}{n+2} < S_n < \frac{2n^2+n}{n^2+1}$

63. C

Sol. $(11-1)^{2023} + (11+1)^{2023} = 11^2.k + {}^{2023}C_1(11+11)$

$\frac{4046}{11} = 367\frac{9}{11}$

64. C

Sol. $M = \frac{5!}{2} = 60$

$N = 120$

65. A

Sol. $\frac{{}^{52}C_2 \times {}^{50}C_1 \times {}^{49}C_1 \times {}^{2}C_1}{{}^{52}C_2 \times {}^{52}C_2 \times {}^{52}C_2}$

66. C

Sol. $|M_n| = n^2 + n - n^2 + 4 = n + 4$

$\sum_{r=1}^{2023} |M_r| = \sum_{r=1}^{2023} r + 4 \times 2023$

67. D

Sol. $(1+x)^{20} = C_0 + C_1x + C_2x^2 + \ldots\ldots + C_{19}x^{19} + C_{20}x^{20}$

$(x-1)^{20} = C_0x^{20} + C_1x^{19} + C_2x^{18} + \ldots\ldots + C_{19}x + C_{20}$

Diff. with respect to x

$20(1+x)^{19} = {}^1C_1 + 2.C_2x + 3C_3x^2 + 4C_4x^3 \ldots\ldots + 20C_{20}x^{19}$ …(i)

$20(1-x)^{19} = {}^{20}C_0x^{19}19C_1x^{18} + \ldots\ldots\ldots - {}^{20}C_{19}$ …(ii)

Equating coeff. of x^{18} in (i) and (ii)

$\sum_{r=0}^{20} (-1)^r \, {}^{20}C_r^2 r(20-r) = -400.{}^{38}C_9$

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68. C

Sol. $r = \sqrt{\frac{9}{4}+1+7} = \frac{\sqrt{41}}{2}$

$AP = \sqrt{8^2+1-24-2-7} = \sqrt{32}$

Area of quadrilateral = $2\left(\frac{1}{2}\times r \times AP\right)$

69. B

Sol. f '(x) = 0 at 2, 3

critical points {0, 2, 3}

f(0) = 5

f(2) = 33

f(3) = 32

70. A

Sol. $\lim\limits_{\alpha\to 0}\left(\frac{e^{\cos\alpha^h}-e}{\alpha^k}\right) = -\frac{e}{2}$ than $\frac{h}{k} = 2$

71. D

Sol. $\frac{dy}{dx}+\frac{e^x}{1+e^x}y = \frac{1}{1+e^x}$

I.F. = $e^{\int\frac{e^x}{1+e^x}dx} = e^{\ln(1+e^x)} = 1+e^x$

$y = (1+e^x) = x + c$

$y = \frac{x+c}{1+e^x}$

As y(0) = 0

So, $y = \frac{x}{1+e^x}$

$y' = \frac{\left(1+e^x\right)-x\left(e^x\right)}{\left(1+e^x\right)^2} = 0$

$e^{-x} = x - 1$

72. C

Sol. $\frac{10-3b^2}{b+2} = -\frac{3\left(b^2-4+4\right)+10}{b+2} = -3(b-2)-\frac{2}{b+2}$ is an integer if $b + 2 = \pm 1$ or ± 2

b = –1, –3, 0, –4

73. A

Sol. $10x^{\sqrt{2}} \le f(x) \le 10x \Rightarrow \left[\int_0^1 f(x)dx\right]$

74. D

Sol. $e = \frac{1}{\sqrt{2}}$

Let $\frac{x^2}{a^2}+\frac{y^2}{a^2\left(1-\frac{1}{2}\right)} = 1$ be the ellipse $x^2 + 2y^2 = a^2$

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75. B

Sol. $\frac{dy}{dx} - y\tan x = \sec x$

$y\cos x = x + c$

$y = \frac{x+3}{\cos x}$

$y' = \frac{\cos x + (x+3)\sin x}{(\cos x)^2}$

76. C

Sol. Equation of normal at (3, 2)

$x - 4y + 5 = 0$

77. C

Sol. Equation of circle is $(x-1)^2 + (y+1)^2 = 12^2$

Point of contact will be $(-11, -1)$

78. D

Sol. $3^x = 4^y = 5^z = 60^{-\omega} = k$

$\frac{\ln 3}{\ln k} = \frac{1}{x}, \frac{\ln 4}{\ln k} = \frac{1}{y}, \frac{\ln 5}{\ln k} = \frac{1}{z}$ and $-\frac{1}{\omega} = \frac{\ln 60}{\ln k}$

79. C

Sol. S.D. $= \left|\frac{(\vec{a}_1 - \vec{a}_2)\cdot(\vec{b}_1 \times \vec{b}_2)}{(\vec{b}_1 \times \vec{b}_2)}\right|$

$\vec{b}_1 \times \vec{b}_2 = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -4 & 3 & 2 \\ -4 & 1 & 1 \end{vmatrix} = \hat{i} - 4\hat{j} + 8\hat{k}$

$\vec{a}_1 - \vec{a}_2 = -\hat{i} - \hat{j}$

S.D. $= \left|\frac{-1+4}{9}\right| = \frac{1}{3}$

80. D

Sol. Roots of f'(x) must be negative and distinct.

SECTION – B

81. 12

Sol. $|\text{adj}(\text{adj}(2A))| = 2^{12}|A|^4$.

82. 4

Sol. $\frac{x-1}{1} = \frac{y+2}{-1} = \frac{z-3}{1} = -2\left(\frac{1+2+3-7}{1+1+1}\right) = \frac{2}{3}$ image of P = Q = $\left(\frac{5}{3}, -\frac{8}{3}, \frac{11}{3}\right)$

$\left|\overrightarrow{OP}\right| = \sqrt{\frac{70}{3}}$

83. 63

Sol. Using LMVT in $[-3, 6]$

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$f'(c) = \dfrac{f(6)-f(-3)}{9}$

$-4 \le f'(c) \le 3$

$-36 \le f(6) - f(-3) \le 27$

$-36 + f(3) \le f(6) \le 27 + f(-3)$

84. 714

Sol. $S_n = \tan^{-1}(n+2) - \tan^{-1}2$

$S_6 = \tan^{-1}\dfrac{6}{17}$

85. 24

Sol. $x\,f(x) - 1 = (x-1)(x-2)(x-3)(x-4)(x-\alpha)$

Put $x = 0$

$\alpha = \dfrac{1}{24}$

$f(x) = \dfrac{(x-1)(x-2)(x-3)(x-4)(x-\alpha)+1}{x}$

$f(5) = 24.$

86. 2

Sol. $f(x) = \int_0^{\pi} \ln\left(1-2x\cos\theta+x^2\right)d\theta$

$f(x) = \dfrac{1}{2}\int_0^{\pi} \ln\left(1+x^4-2x^2\cos 2\theta\right)d\theta$

$f(x) = \dfrac{1}{4}\int_0^{2\pi} \ln\left(1+x^4-2x^2\cos\theta\right)d\theta$

$f(x) = \dfrac{1}{2}\int_0^{\pi} \ln\left(1+x^4-2x^2\cos\theta\right)d\theta = \dfrac{f\left(x^2\right)}{2}$

$\dfrac{f\left(x^2\right)}{f(x)} = 2$

87. 6

Sol. Locus of z is circle

i.e. $x^2 + y^2 = 1$

88. 81

Sol. $12x^6 + 2x^3 - 10 = 0$

$x^3 = \dfrac{5}{6}, -1$

89. 24

Sol. $\left(2\vec{a}+\vec{b}+\vec{c}\right)\cdot\left(\left(\vec{a}\cdot\vec{b}\right)-\left(\vec{b}\cdot\vec{b}\right)\vec{a}+6\vec{c}\right)$

$\left(2\vec{a}+\vec{b}+\vec{c}\right)\cdot\left(-9\vec{a}+6\vec{c}\right)$

$-18|\vec{a}|^2 + 6|\vec{c}|^2$

$-72 + 6\times 16 = 24$

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90. 5

Sol. Let tanD is fourth root

$$\tan(A + B + C + D) = \frac{\Sigma \tan A - \Sigma \tan A \cdot \tan B \cdot \tan C}{1 - \Sigma \tan A \cdot \tan B + \Pi \tan A} = \frac{5}{3}$$

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